

Precision & Recall

A Complete Guide to Two Core Classification Metrics

When evaluating a classification model, raw accuracy often tells an incomplete story — especially when one class is far rarer than the other. **Precision** and **Recall** cut through that noise, each measuring a different dimension of your model's ability to identify the positive class correctly and completely.

Metric	Core Question	Range	Hurt By
Precision	Are my Positive predictions correct?	0 – 1	False Positives (FP)
Recall	Did I find ALL the real Positives?	0 – 1	False Negatives (FN)

Table — At-a-glance summary

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1. Precision

Precision measures how trustworthy the model's positive predictions are. When the model says "this is a positive case," how often is it actually right? It is the metric of **exactness** — a high-precision model rarely raises false alarms.

Formula

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Where **TP** (True Positives) are correct positive predictions and **FP** (False Positives) are incorrect positive predictions. The denominator is simply every instance the model predicted as positive.

Plain-English Interpretation

Imagine the model as a detective filing arrest warrants. Precision = the fraction of issued warrants that result in a valid conviction. A low-precision detective arrests too many innocent people.

Real-World Analogies

Spam Filter	Precision = fraction of emails flagged as spam that are genuinely spam. Low precision means legitimate emails get blocked — users lose trust.
Search Engine	Precision@10 = fraction of the top 10 results that are truly relevant. Low precision forces users to scroll past irrelevant links.
Gold Prospector	Precision = fraction of rocks brought back that actually contain gold. Low precision wastes effort transporting worthless material.

Worked Numeric Example

A medical imaging AI scans 500 X-rays and flags **80** as showing pneumonia. After expert review, **64** of those 80 are confirmed cases and **16** are not.

- True Positives (TP) = 64
- False Positives (FP) = 16
- Precision = $64 / (64 + 16) = 64 / 80 = \mathbf{0.80}$

The model is correct 80% of the time it raises an alert — one in five alerts is a false alarm.

What Lowers Precision?

Cause	Why It Hurts Precision
Low decision threshold	Predicts Positive too liberally → more FP
Noisy/irrelevant features	Model confuses negatives for positives
Severe class imbalance	Large negative pool creates more FP opportunities
Overly broad boundary	Positive region captures many true negatives

Table 1 — Causes of low Precision

How to improve Precision: Raise the decision threshold • Add stronger/cleaner features • Apply higher misclassification cost on FP during training • Use precision-oriented resampling strategies.

2. Recall

Recall measures how completely the model finds all real positive cases. Of every actual positive that exists in the data, how many did the model catch? It is the metric of **completeness** — a high-recall model misses very few real cases. Also known as **Sensitivity** or **True Positive Rate (TPR)**.

Formula

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Where **FN** (False Negatives) are actual positives the model failed to detect. The denominator is the total number of real positives in the dataset.

Plain-English Interpretation

Continuing the detective analogy: Recall = the fraction of all real criminals that the detective actually caught. Low recall means guilty parties walk free.

Real-World Analogies

Cancer Screening	Recall = fraction of all cancer patients correctly identified. Low recall means sick patients are sent home without treatment.
Fraud Detection	Recall = fraction of all fraudulent transactions that get flagged. Low recall means fraudsters slip through undetected.
Search & Rescue	Recall = fraction of all missing survivors that are found. Low recall means some people are never located.

Worked Numeric Example

There are **100** actual pneumonia cases across the 500 X-rays. The AI found **64** of them and missed **36**.

- True Positives (TP) = 64
- False Negatives (FN) = 36
- Recall = $64 / (64 + 36) = 64 / 100 = \mathbf{0.64}$

The model detected only 64% of real pneumonia cases — 36 patients were incorrectly cleared.

What Lowers Recall?

Cause	Why It Hurts Recall
High decision threshold	Predicts Positive too conservatively → more FN
Positive class under-represented	Model rarely seen enough positive examples to learn from
Severe class imbalance	Minority class is "drowned out" during training
Overly conservative boundary	Model prefers "safe" negative predictions

Table 2 — Causes of low Recall

How to improve Recall: Lower the decision threshold • Oversample positives (SMOTE) • Assign higher misclassification cost to FN • Use ensemble methods that boost sensitivity on rare classes.

3. The Precision–Recall Trade-off

Precision and Recall are locked in tension. Adjusting the classification threshold moves one metric up while pulling the other down. This is one of the most important dynamics in applied machine learning.

How the Threshold Drives Both Metrics

Threshold	Behaviour	Precision	Recall
Very High (≥ 0.95)	Predict Positive only when near-certain	↑ High	↓ Low
Default (0.50)	Standard balanced prediction	Moderate	Moderate
Very Low (≤ 0.10)	Predict Positive very liberally	↓ Low	↑ High

Table 3 — Threshold direction and its effect on each metric

Why the Trade-off Exists

- Lowering the threshold makes the model predict Positive more often → catches more real positives (**Recall** ↑) but also flags more negatives as positive (**Precision** ↓).
- Raising the threshold makes the model predict Positive only when very confident → fewer false alarms (**Precision** ↑) but some real positives are now missed (**Recall** ↓).
- There is no free lunch: you cannot simultaneously maximise both metrics with a single threshold.

Threshold Selection in Practice

Goal	Strategy
Maximise F1 Score	Find threshold where Precision $\approx$ Recall on validation set
Minimise false alarms	Raise threshold until Precision meets the business SLA
Maximise coverage	Lower threshold until Recall meets the clinical requirement
Cost-sensitive decision	Sweep threshold; minimise: $\text{cost_FP} \times \text{FP} + \text{cost_FN} \times \text{FN}$

Table 4 — Choosing the right threshold

4. The Precision–Recall Curve & AUPRC

Rather than evaluating at a single threshold, the **Precision–Recall Curve** plots Precision (y-axis) against Recall (x-axis) across every possible threshold. This gives a full picture of the trade-off space.

Sample PR Curve Values

Threshold	Precision	Recall	Observation
0.95	0.98	0.30	Very safe — misses most positives
0.80	0.93	0.55	High precision, moderate recall
0.50	0.80	0.80	Balanced operating point
0.30	0.60	0.91	Good recall, noisier predictions
0.10	0.32	0.98	Catches almost everything — lots of FP

Table 5 — Example PR Curve values at different thresholds

Area Under the PR Curve (AUPRC)

The AUPRC summarises the entire curve as one number (0–1). Higher is better. It is especially valuable on imbalanced datasets because it focuses entirely on the positive class and is not inflated by the large number of true negatives — unlike AUC-ROC.

Rule of thumb: A random classifier on a dataset with 10% positive class has an AUPRC $\approx$ 0.10. A useful model should significantly exceed the positive class prevalence. Use AUPRC whenever classes are highly imbalanced.

5. Numeric Model Comparison

Four classifiers evaluated on 1 000 samples (100 positive, 900 negative):

Model	TP	FP	FN	Precision	Recall	Verdict
Aggressive	95	190	5	0.33	0.95	High recall, low precision
Balanced	80	20	20	0.80	0.80	Solid on both
Conservative	50	5	50	0.91	0.50	High precision, low recall
Near-perfect	97	8	3	0.92	0.97	Strong on both

Table 6 — Four models compared (red = weak, green = strong)

6. When to Prioritise Each Metric

Prioritise PRECISION When...	Prioritise RECALL When...
False alarms are costly (spam blocking real mail)	Missing a positive is dangerous (cancer diagnosis)
User trust depends on accuracy (recommendation engines)	Coverage is the primary goal (content moderation)
Resources to act on alerts are scarce (security analyst capacity)	All positives MUST be found (regulatory compliance)
Wrong positive = high financial cost (expensive lab test ordered)	Wrong negative = high human cost (disease untreated)

Table 7 — Decision guide for choosing your primary metric

Python Code


```
from sklearn.metrics import precision_score, recall_score, precision_recall_curve

y_true = [0, 1, 1, 0, 1, 1, 0, 0, 1, 0]
y_pred = [0, 1, 0, 0, 1, 1, 1, 0, 1, 0]
y_prob = [.1, .9, .4, .2, .8, .7, .6, .3, .85, .15]

# Point metrics at default 0.5 threshold
p = precision_score(y_true, y_pred)
r = recall_score(y_true, y_pred)
print(f"Precision: {p:.3f} | Recall: {r:.3f}")

# Full PR Curve (threshold-independent)
prec, rec, thresholds = precision_recall_curve(y_true, y_prob)
```

Key Takeaways

- **Precision** = $TP / (TP + FP)$ — exactness; "when I say positive, am I right?"
- **Recall** = $TP / (TP + FN)$ — completeness; "did I find all the real positives?"
- Raising threshold → Precision ↑, Recall ↓. Lowering → Precision ↓, Recall ↑.
- No threshold maximises both simultaneously — choose based on your cost of errors.
- Use the PR Curve and AUPRC for threshold-independent model comparison.
- Combine into F1 (or F-beta) when you need a single balanced summary.